This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scripter Modules, refer to the "Guide to Using Scripter Modules" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): CP2210, CP2215, CP2220, CP2230, CP2000-ZX, CP2000-M

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scripter Version
3.13.0000-b006	2.10.0

Version History

Module Version	Date	Notes
1_1_3_0	7/8/2021	Fixed the following commands: Shutter, Power, Video Mute and Channel. Fixed status feedback and removed Warming Up and Cooling Down states from Power command.
1_0_2_0	4/19/2017	Added Christie IMB Standby Mode & Full Power Mode states to Power command. Fixed import error.
1_0_1_0	2/1/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
  `Example: InterfaceName.Unidirectional = 'True'`

- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
  `Example: InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass

```
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='CP2210')
```

SerialOverEthernetClass

```
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='CP2210')
```

EthernetClass

```
InterfaceName = ModuleName.EthernetClass('192.168.254.254', 3002, Model='CP2210')
```

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	Value
AutoImage	None		
`# AutoImage example` `InterfaceName.Set('AutoImage', None)`			
Channel	'1' – '64'		
`# Channel example` `InterfaceName.Set('Channel', '1')`			
Focus	'Positive'	'Negative'	'Stop'
`# Focus example` `InterfaceName.Set('Focus', 'Positive')`			
Input [1]	'292-A' 'DVI-A'	'292-B' 'DVI-B'	'292-Dual' 'DVI-Dual/Twin'
`# Input example` `InterfaceName.Set('Input', '292-A')`			
LensShiftHorizontal	'Positive'	'Negative'	'Stop'
`# LensShiftHorizontal example` `InterfaceName.Set('LensShiftHorizontal', 'Positive')`			
LensShiftVertical	'Positive'	'Negative'	'Stop'
`# LensShiftVertical example` `InterfaceName.Set('LensShiftVertical', 'Positive')`			
Power	'On' 'Full Power Mode'	'Off'	'Christie IMB Standby Mode'
`# Power example` `InterfaceName.Set('Power', 'On')`			
VideoMute	'On'	'Off'	
`# VideoMute example` `InterfaceName.Set('VideoMute', 'On')`			
Zoom	'Positive'	'Negative'	'Stop'
`# Zoom example` `InterfaceName.Set('Zoom', 'Positive')`			

[1] Status from 'Channel' command is required for this command to work.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
	InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
	Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
	InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, 
FeedbackHandler)
	FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
	InterfaceName.Update(Command)
	Value = InterfaceName.ReadStatus(Command)
	InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
	FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
Channel	'1' – '64'		

```
# Channel example
InterfaceName.Update('Channel')
Value = InterfaceName.ReadStatus('Channel')
InterfaceName.SubscribeStatus('Channel', None, FeedbackHandler)
```

Command	Value	Value	Value
ConnectionStatus	'Connected'	'Disconnected'	

```
# ConnectionStatus example
Value = InterfaceName.ReadStatus('ConnectionStatus')
InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)
```

Command	Value	Value	Value
Input	'292-A'	'292-B'	'292-Dual'
	'DVI-A'	'DVI-B'	'DVI-Dual/Twin'

```
# Input example
InterfaceName.Update('Input')
Value = InterfaceName.ReadStatus('Input')
InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)
```

Command	Value	Value	Value
LampUsage	Hours		

```
# LampUsage example
InterfaceName.Update('LampUsage')
Value = InterfaceName.ReadStatus('LampUsage')
InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)
```

Command	Value	Value	Value
Power	'On'	'Off'	'Christie IMB Standby Mode'
	'Full Power Mode'		

```
# Power example
InterfaceName.Update('Power')
Value = InterfaceName.ReadStatus('Power')
InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)
```

Command	Value	Value	Value
VideoMute	'On'	'Off'	

```
# VideoMute example
InterfaceName.Update('VideoMute')
Value = InterfaceName.ReadStatus('VideoMute')
InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)
```

Cable and Adapter Requirements

Captive Screw to Male DB9 RS232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232, RS-422

Baud Rate: 115200

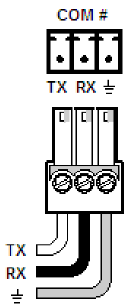
Data Bits: 8

Parity: None

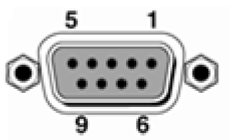
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



RS232 DB9, Female Projector

Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface.

Port Type:	Ethernet (TCP)
Default Port:	3002
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Auto Image None	(LLM+AUTO 1)
Channel 1	(CHA 101)
Channel 64	(CHA 164)
Focus Negative	(FCS+STRT -1)
Focus Positive	(FCS+STRT 1)
Focus Stop	(FCS+STOP)
Input 292-A (Channel 1)	(SIN+C101 1)
Input 292-B (Channel 1)	(SIN+C101 2)
Input 292-Dual (Channel 1)	(SIN+C101 3)
Input DVI-A (Channel 1)	(SIN+C101 4)
Input DVI-B (Channel 1)	(SIN+C101 5)
Input DVI-Dual/Twin (Channel 1)	(SIN+C101 6)
Lens Shift Horizontal Negative	(LHO+STRT -1)
Lens Shift Horizontal Positive	(LHO+STRT 1)
Lens Shift Horizontal Stop	(LHO+STOP)
Lens Shift Vertical Negative	(LVO+STRT -1)
Lens Shift Vertical Positive	(LVO+STRT 1)
Lens Shift Vertical Stop	(LVO+STOP)
Power Christie IMB Standby Mode	(PWR2)
Power Full Power Mode	(PWR1)
Power Off	(PWR3)
Power On	(PWR0)
Video Mute Off	(SHU 1)
Video Mute On	(SHU 0)
Zoom Negative	(ZOM+STRT -1)
Zoom Positive	(ZOM+STRT 1)
Zoom Stop	(ZOM+STOP)

Appendix B. Update Commands

Channel	`(CHA?)`
Input	`(SIN?)`
Lamp Usage	`(LPH?)`
Power	`(PWR?)`
Video Mute	`(SHU?)`